OpenClaw RGB LED Control Tutorial

> Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Switching Models], [04-AI Voice Interaction Configuration], [05-Three Chat Modes Configuration Testing]. If you are currently only using TUI / WhatsApp, you can skip `04` for now. This document assumes the basic environment is already prepared and only covers RGB-related content.

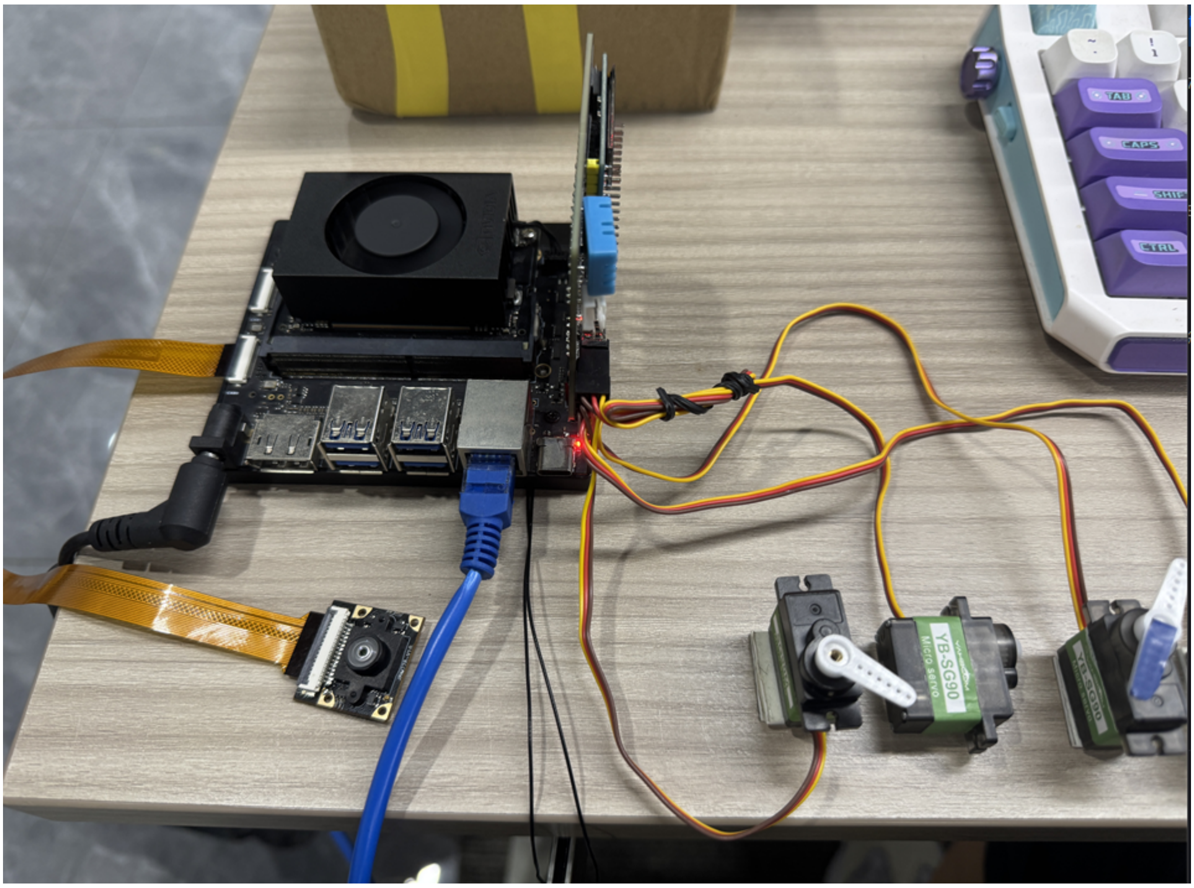
1. Tutorial Objectives

This tutorial demonstrates OpenClaw's natural language control capability through RGB LED peripherals. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw controlling RGB LEDs
- Chat with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four RGB examples: single color lighting, breathing light, warning flash, rainbow marquee
- Understand the complete pipeline from "user input" to "light execution"

2. Hardware Preparation

- OpenClaw main control board
- RGB LED strip / RGB LED module `[default 14 WS2812 LEDs]`
- `[Optional] Microphone module`
- Wiring diagram



You can add your own RGB wiring diagram and hardware photo here.

3. Software Preparation

Code entry points used in this tutorial:

- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- RGB control command: `/home/jetson/hardware_hal_demo/rgb`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The `rgb` tool already supports the following capabilities:

- Set fixed color `color`
- Set light effect `effect`
- Turn off lights `off`

Commonly used light effects include:

- `breathe` breathing light
- `blink` flash
- `flow` flowing light
- `marquee` marquee light
- `rainbow` rainbow light

If you want to individually verify the low-level RGB commands, you can also enter the following in the terminal:

```
rgb color --color red  # All lights turn red
rgb effect --effect breathe --color blue --speed 5   # Breathing light
rgb off   # Turn off lights
```

For more commands, see the markdown documentation in this folder:

```
/home/jetson/hardware_hal_demo/README.md
```

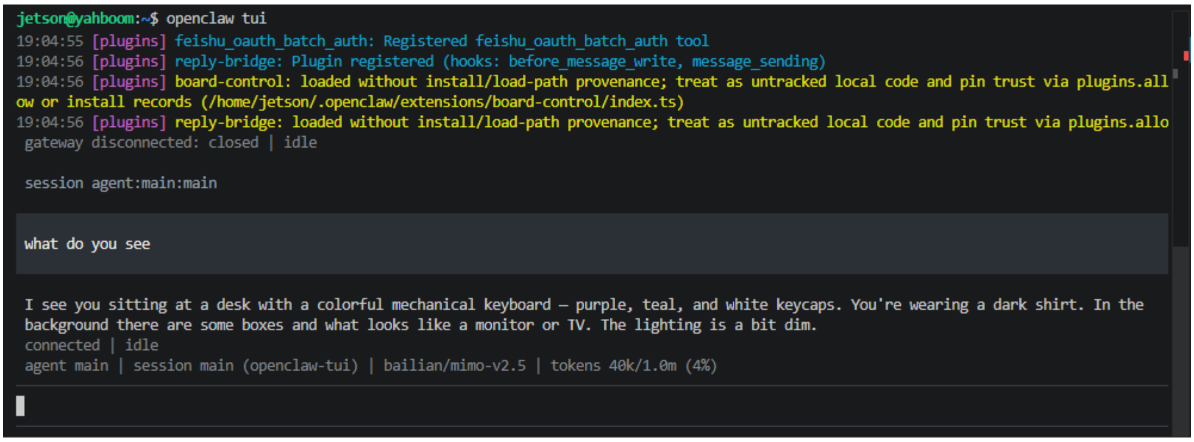
If the low-level commands can execute normally, it means the hardware connection to the Jetson Orin NX is correct and communication is established.

4. Testing RGB LEDs

Before starting this tutorial, please complete a basic `openclaw tui` chat self-check; if you haven't done so, see [05-Three Chat Modes Configuration Testing] first. After confirming OpenClaw can reply normally, proceed to the RGB-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```



You can try entering the following in the terminal:

- Set all RGB LEDs to red
- Change the RGB LEDs to blue
- Lower the RGB LED brightness a bit
- Turn off the RGB LEDs

If you want to test individual LEDs, you can also try entering content like:

- Set LED 1 to green
- Set LED 5 to purple

5. Three Methods to Chat with OpenClaw

Method	Suitable Scenarios	Advantages	Recommended Positioning
WhatsApp	Remote control, classroom demo	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First get the control pipeline working
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration method

If the AI voice module + speaker is an optional add-on, Feishu chat, TUI, and WhatsApp chat need to play back OpenClaw's replies. You can run the following program in the terminal. AI voice module chat does not need to run it again.

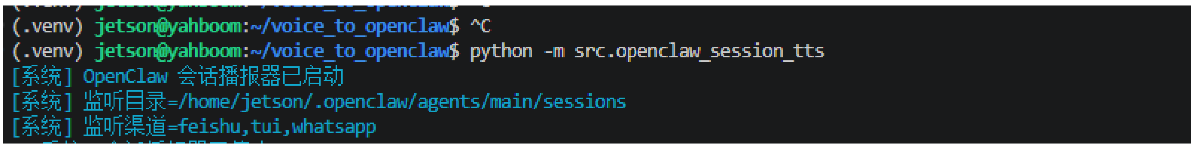
- Modify the broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether to broadcast. After modification, save and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

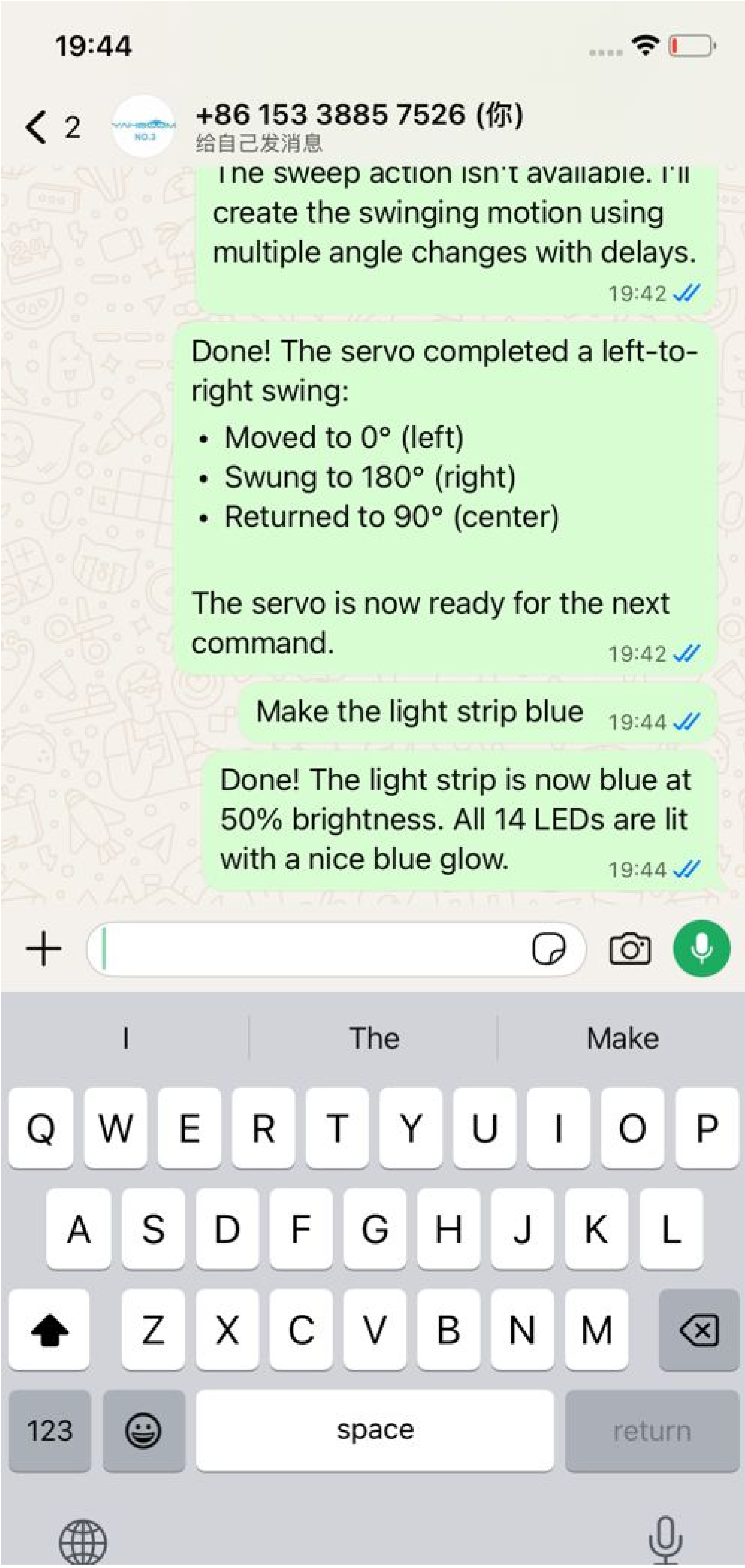


5.1 whatsapp Method

For WhatsApp connection steps, please refer directly to [05-Three Chat Modes Configuration Testing]. This section only retains chat examples suitable for the RGB tutorial.

After creation, you can chat directly, for example:

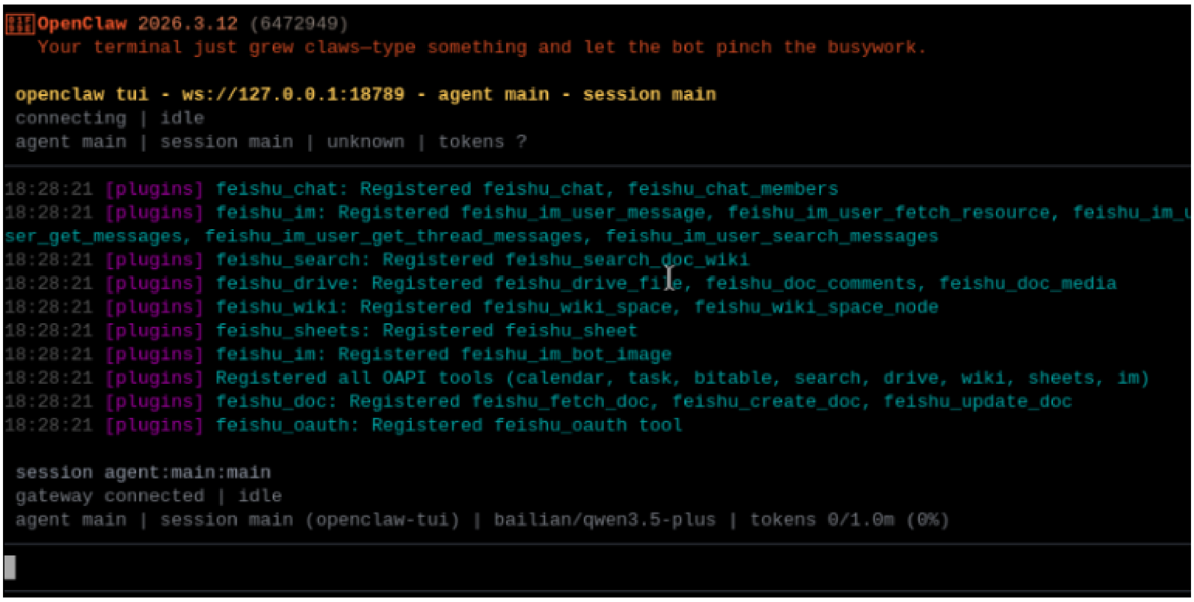
- Change the LED strip to blue
- Turn on the rainbow light effect
- Make the lights flash faster


19:44
2
+86 153 3885 7526 (你)
给自己发消息
The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.
19:42
Done! The servo completed a left-to-right swing:
• Moved to 0° (left)
• Swung to 180° (right)
• Returned to 90° (center)
The servo is now ready for the next command.
19:42
Make the light strip blue
19:44
Done! The light strip is now blue at 50% brightness. All 14 LEDs are lit with a nice blue glow.
19:44
I
The
Make
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
space
return

5.2 TUI Method

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue chatting:

```
openclaw tui
```

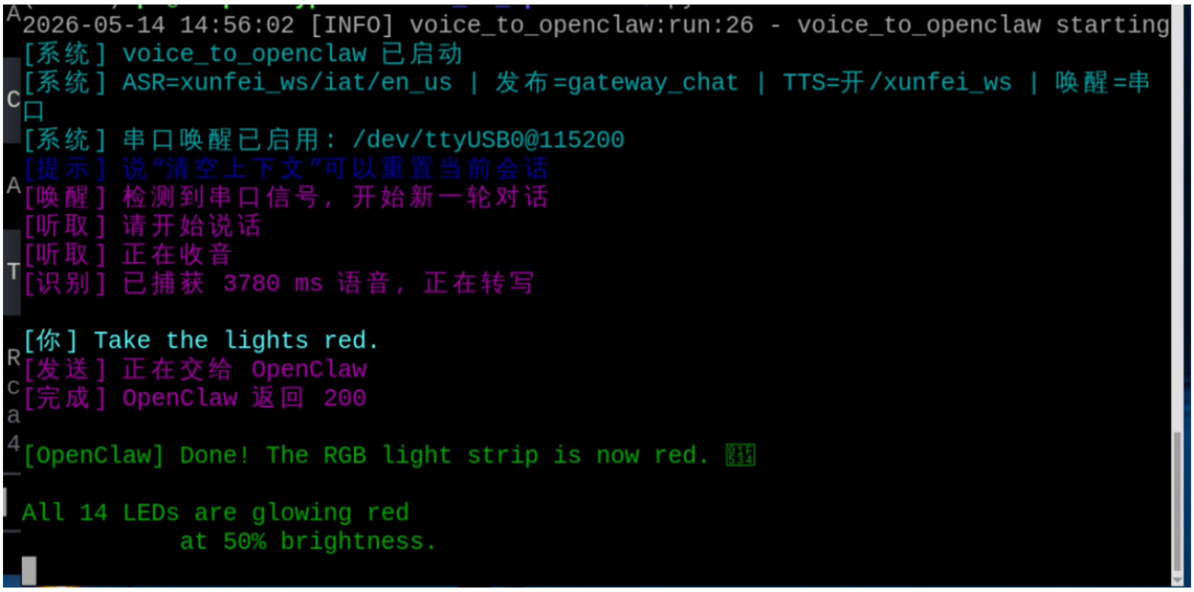


5.3 Voice Code Method

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake process, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can speak your control commands directly, for example:

- Change the lights to red
- Turn on the blue breathing light
- Make the lights flow like a rainbow



6. Comprehensive Cases

Note: The following are all demonstrated using `openclaw tui` for chat. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Single Color Lighting

Experimental Objective

Make the RGB LEDs consistently display a single fixed color, completing the most basic light control.

Effect Description

After receiving the command, all RGB LEDs uniformly display the specified color, such as red, green, blue, white, or warm white. This case is suitable for verifying that color mapping is working correctly.

Example Commands

- Set all RGB LEDs to red
- Change the LED strip to blue
- Change the lights to warm white

6.2 Case 2: Breathing Light

Experimental Objective

Make the RGB LEDs gradually vary in brightness on a given color, creating a breathing light effect.

Effect Description

The RGB LEDs will cycle from dim to bright, then from bright to dim. The overall effect is relatively soft, suitable for standby indicators, ambient lights, or status indicator lights.

Example Commands

- Turn on the blue breathing light
- Change the lights to a green breathing effect
- Slow down the breathing light speed a bit

6.3 Case 3: Warning Flash

Experimental Objective

Make the RGB LEDs rapidly flash to simulate a warning light or alarm effect.

Effect Description

The lights can flash rapidly using red, yellow, or alternating red and blue, suitable for expressing information like "attention", "warning", or "abnormal state", with more obvious visual feedback.

Example Commands

- Make the lights flash red
- Turn on the yellow warning light
- Make the RGB LEDs flash rapidly three times

6.4 Case 4: Rainbow Marquee

Experimental Objective

Make the RGB LEDs form a continuously flowing dynamic effect, demonstrating richer light expression.

Effect Description

The LED colors will change sequentially and flow in one direction, looking like a rainbow flowing light or marquee light. This effect is very suitable for boot animations, demonstration animations, or interactive light effects.

Example Commands

- Turn on the rainbow light effect
- Make the lights flow like a marquee
- Turn on the blue flowing light, with a faster speed